

# Gigabit Ethernet

Module 25

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Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar University of Science and Technology Hisar harana I welcome you all to the video lecture series of cyber and information security in this lecture we will study about gigabit Ethernet which is basically used for data transmission on wired Lan at speed of 1,000 Mbps or 1 gbps first of all please see the contents which I shall cover in this lecture after covering brief introduction about gigabit Ethernet I shall take up layer protocol architecture of gigabit Ethernet which will be followed by gigabit Ethernet physical specifications in last I shall take up frame format of gigabit Ethernet so now let us start one of the basic laws of network

design is that we always underestimate the demand for increased capacity only a few years ago the idea of an Ethernet running at gigabit speeds would have seemed to be a little excessive but today's data intensive applications and increasing number of network users and new methods of information delivery are driving an ever increasing demand for more bandwidth when 100 Mbps Network working Technologies like fddi were first introduced most horizontal networks used 10 Mbps ethernet these new protocols were used primarily on backbones now that fast ethernet has taken over the horizontal Network Market a 100 Mbps backbone is in many cases insufficient to support the connections between switches that have to

accommodate multiple fast ethernet networks gigabit Ethernet was developed to be the next generation of ethernet Network running at 1 gbps or 1000 Mbps speed that is 10 times the speed of fast ethernet now let us see i e standards for gigabit Ethernet the initial standard for gigabit Ethernet was produced by i e in June 1998 as i e 802.3 Zed and it required optical fiber for communication 802.3 Z is commonly referred to as thousand base X where X refers to either CX SX LX or non standard ZX i e 802.3ab ratified in 1999 defines gigabit Ethernet transmission over unshielded twisted pair UTP Cat 5 5e or six cable known as thousand base t with the rectification of 802.3 ab gigabit Ethernet became a

desktop technology as organizations could use their existing copper cabling infrastructure i.e. 802.3ah ratified in 2004 added two more gigabit Ethernet fiber standards thousand base LX 10 and 1,000 base bx10 this was part of a larger group of protocols known as ethernet in the first mile although it is still a relatively new technology gigabit Ethernet has virtually assured a place in the market like Fast ethernet it uses the same frame format frame size and media control method as used in standard 10 Mbps ethernet fast ethernet has overtaken fddi because it prevented Network administrators from having to use a different protocol on the backbone in the same way gigabit Ethernet prevents administrators from having to

use a different protocol like ATM for their backbones like all other ethernet Technologies the gigabit Ethernet standard is an extension of the base I 802.3 standard the specific design objectives for the gigabit Ethernet specification given to the task force includes the following should offer 10 times the bandwidth of fast ethernet must use the i.e. 802.3 ethernet frame format should employ the same half duplex and full duplex Mac operation schemes as its predecessor should be backward compatible with 10 Mbps and 100 Mbps ethernet Technologies should support all existing Network protocols used with the e ethernet family some of the more important features of gigabit Ethernet are outlined here supports the existing frame size and frame format of I 802.3

ethernet which means it easily integrates with existing networks allows an easy upgrade path to High Performance networking while taking 7 minutes, 1 second advantage of existing Technologies and product knowledge supports full duplex mode for switch to switch and switch to end station connections most gigabit Ethernet products being shipped are full duplex in this mode there are no shared medium contention problems support s half duplex mode on shared network connections in this mode gigabit Ethernet use the same fundamental CSM by CD medium access method and traditional ethernet to resolve contention for the shared media a packet burst in feature was added to csma by CD that allows servers

switches and other devices to send burst of small packets in order to fully utilize available bandwidth uses fiber optic cable or Cat 5 or cat 6 cable one of the main benefits of gigabit standard is that it offers a lowcost solution to solve the demands for bandwidth not only the cost of installation is low but the cost of network maintenance and management is minimum as well management and maintenance for gigabit Ethernet may be done by local network administrators in addition to the cost reduction benefit gigabit Ethernet may allow faster switching since gigabit Ethernet uses the same frame format it allows seamless integration of Lan man and when there is no need for packet fragmentation reassembling or address translation

eliminating the need for routers that are much slower than switches now let us see layer protocol architecture of gigabit ethernet gigabit Ethernet supports 10 100 and 1,000 Mbps data rates it provides separate 8 bit wide receive and transmit data paths so it can support both full duplex as well as half duplex operation the GMI provides two media status signals

one indicates presence of the carrier and the other indicates absence of collision the various layers of the gigabit Ethernet protocol architecture are shown in figure the gmii is the interface between the Mac layer and the physical layer it allows any physical layer to be used with the Mac layer it is an extension of the Mii media independent dependent

interface used in Fast ethernet it uses the same management interface as Mii with the gmii it is possible to connect various media types such as shielded and unshielded twisted pair and single mode and multimode optical fiber while using the same Mac controller the GMI sits above three main sub layers the PCS the PMA and the PMD let us see each layer one by one PCS this layer of the GMI provides a uniform interface for all possible physical media it uses an 8B by 10B coding scheme just like fiber channel in this type of coding 10bit code groups represent group of eight bits the PCS sublayer also generates the carrier sense and collision detection indications for half duplex operation it manages the auto negotiation process by which a NIC communicates with the

network to determine the network speed that is 10 100 or 1000 Mbps and mode of operation as half duplex or full duplex PMA this sublayer provides a medium independent means for the PCS to support various serial bit oriented physical media this layer takes the 10bit code groups sent at 125 mhz by the PCS and converts them into serial format before Transmission in the reverse Direction it deserializes bits received from transmitter back into code groups for delivery to the PCS PMD this sublayer Maps the physical medium to the PCS this layer defines the physical layer signaling used for various media the medium dependent interface MDI a component of the PMD is the actual physical layer interface this layer defines the actual physical attachment such as connectors

for different media types it defines different type of physical standards like 802.3 Z 802.3ab and 802.3 a now after layer protocol architecture let us see physical specifications of gigabit Ethernet the physical layer of gigabit Ethernet describes the physical properties of the communication media as well as the electrical properties and interpretation of the exchanged signals following three types of media specified in the i.e 802.3 z standard collectively known as thousand base X standard are as shown in table thousand base SX it is a fiber optic gigabit Ethernet standard it operates over multimode fiber using a 850 nanom near infrared light wavelength the specification allows for a maximum distance between end points of 220 M over

62.5 or 125 microm fiber with good quality Terminators thousand base SX will usually work over significantly longer distances modern 50 or 125 micrometer fiber can reliably extend the signal to 500 M or more this standard is highly popular for intra building links in large Office Buildings collocation facilities and carrier neutral internet exchanges thousand base LX it is a fiber optic gigabit Ethernet standard using a long wavelength laser its wavelength is 1270 to 135 nanometer typically gigabit Ethernet lasers will be specified as having a 1300 or 1310 nanom wavelength thousand base LX is specified to work over a distance of up to 5 kilom over 10 microm single mode fiber in practice it will often operate correctly over a much

greater distance many manufacturer will guarantee operations up to 10 or 20 kilom provided that their equipment is used at both ends of the link for any link distance greater than 300 M the use of a special launch conditioning patch cord may be required this launches the laser at a precise offset from the center of the fiber which causes it to spread across the diameter of the fiber core base CX it is an initial standard for gigabit Ethernet connections with maximum distance of 25 M using balanced shielded twisted pair with a pinout different from, base T the short segment length is due to very high signal transmission rate although it is still used for specific applications for example ethernet connections between the blade servers and the switch modules but

now thousand base T has succeeded it for Journal copper wiring use physical specifications under i e 802.3ab is, base t or TX as shown in table th000 Base TX the telecommunication industry Association TI created and promoted a version of thousand based T that was simpler to implement calling it th000 Base TX the simplified design would in theory have reduced the cost of the required Electronics by using only two pair in each Direction many thousand based T products are advertised as thousand based TX due to lack of knowledge that thousand Base DX is actually a different standard physical specifications under i e 802.3 AP also known as ethernet in the first mile is, base KX th000 base lx10 and th000 base bx10

as shown in table th000 base KX it is part of the i e 802.3 AP standard for ethernet operation over electrical back planes this standard defines one to four lanes of back plane links one RX and one TX differential pair per Lane at link bandwidth ranging from megabit to 10 gabit per second, base lx10 it was standardized 6 years after the initial gigabit fiber version as part of the ethernet in the first mile task group it is very similar to, base LX but achieves longer distances up to 10 km over a pair of 1310 nanometer wavelength single mode fiber due to higher quality Optics thousand base bx10 it is capable of up to 10 kilom over a single strand of single mode fiber with a different wavelength going in each Direction the terminals on each

side of the fiber are not equal as the transmitting Downstream that is from the center of the network to the outside uses the 1490 nanometer wavelength and the one transmitting Upstream uses the 1310 nanometer wavelength few other known standard physical specifications are thousand base ex and thousand base ZX as shown in table thousand base ex is a nonn standard but industry accepted term to refer to gigabit Ethernet transmission it is very similar to th base lx10 but achieves longer distances up to 40 kilom over a pair of single mode fiber due to higher quality Optics then lx10 running on 1310 nanometer wavelength lasers thousand base ZX is a known standard but multivendor term to refer to gabit ethernet transmission

using 1550 nanometer wavelength to achieve distances of at least 70 kilom over single mode fiber some vendors specify distances up to 120 kilom over single mode fiber sometimes called thousand base EZ X essentially four types of Hardware are needed to upgrade an existing ethernet or fast ethernet Network to gigabit Ethernet gigabit Ethernet network interface card

Nic aggregating switches that connect a number of fast ethernet segments 2 gab ethernet gigabit Ethernet switches gigabit Ethernet repeaters now let us see CSM by CD and frame format of gigabit Ethernet the Mac layer of gigabit Ethernet uses the same csma by CD protocol as used in Ethernet the maximum length of a cable segment used to

connect stations is limited by the CSM ycd protocol if two stations simultaneously detect an idle medium and start transmitting a collision occurs the i e 802.3 z Mac operation can be either half or full duplex mode a half duplex channel can receive and transmit but not at the same time with full duplex transmission it is possible to transmit and receive data at the same time in full duplex mode the gigabit Ethernet Mac uses the i e 802.3x full duplex specification which includes the i e 802.3x flow control in half duplex mode the gigabit Ethernet Mac uses the long established CSM by CD assess method the use of full duplex Transmission in gabit ethernet increases the overall bandwidth from 1 to 2 gbps for point2 Point links it also increases the maximum

transmission distances for the particular media CSM by CD is not used for media access in full duplex gigabit Ethernet because the use of full duplex operation eliminates collisions on The Wire full duplex operation is best suited to Backbone transmission and as assess to highspeed servers the protocol supports both half and full duplex mode of operation an enhancement that was introduced in the switched version of 100 Mbps ethernet in the half duplex mode of operation the protocol maintains the CSM by CD feature of the original protocol now coming on to gigabit Ethernet frame format the gigabit Ethernet defines a basic data frame format that is required for all Mac implementations plus several additional optional formats that are used to extend

the protocol's basic capability the basic data frame format contains the Seven Fields as shown in figure Preamble PR it consists of seven bytes the Preamble is an alternating pattern of ones and zeros that tells receiving stations that a frame is coming and that provides a means to synchronize the frame reception portion of receiving physical layers with the incoming bit stream start of frame s soof it consist of one byte the soff is an alternating pattern of ones and zeros ending with two consecutive one bits indicating that the next bit is the leftmost bit in the leftmost most bite of the destination address destination address da consist of six bytes the da field identifies which station should receive the frame the leftmost bit in the da field indicates whether the address is an

individual address which is indicated by a zero or a group address which is indicated by a one the second bit from the left indicates whether the da is globally administered which is indicated by a zero or locally administered which is indicated by a one the remaining 46 bits are a uniquely assigned value that identifies a single station a defined group of stations or all stations on the Network Source address sa consist of six bytes the sa field identifies the sending station the essay is always an individual address and the leftmost bit in the essay field is always zero length oblique type it consist of two bytes this field indicates either the number

of Mac clients data bytes that are contained in the data field of

the frame or the frame type ID if the frame is assembled using an optional format if the length or type field value is less than or equal to 1500 the number of LLC bytes in the data field is equal to the length or type field value if the length type field value is greater than 1500 the frame is an optional type frame and the length type field value identifies the particular type of frame being sent or received data it is a sequence of n bytes of any value where n is less than or equal to 1,500 if the length of the data field is less than 46 the data field must be extended by adding a filler or a pad sufficient to bring the data field length to 46 bytes frame check sequence FCS it consists of four bytes this sequence contains a 32bit cyclic redundancy check CRC value

which is created by the sending Mac and is recalculated by the receiving MEC to check for damaged frames the FCS is generated over the data length type and data fields extension gigabit Ethernet should be interoperable with existing 802.3 networks carrier extension is a way of maintaining i.e. 802.3 minimum and maximum frame sizes with meaningful cable distances for Carrier extended frame the known data extension symbols are included in the Collision window that is the entire extended frame is considered for collision and dropped however the frame check sequence FCS is calculated only on the original frame the extension symbols are removed 31 minutes, 1 second before the FCS is checked by the receiver so friends this was gigabit

Ethernet finally summarizing we learned about introduction to gigabit Ethernet which was followed by layered protocol architecture of gigabit Ethernet and then gigabit Ethernet physical specifications in the end we discussed frame structure of gigabit Ethernet hope the contents explained in this lecture were understandable and helpful hope to see you in the next lecture till then goodbye enjoy the day thank you